

Last Time:

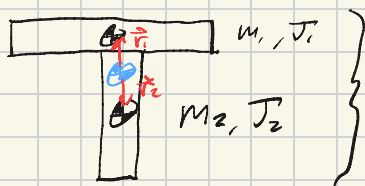
- Finished MEMS byro
- Rigid Bodies
- Inertia Matrix
- Kinetic Energy

Today:

- Parallel-Axis Theorem
- Momentum
- Euler's Equation

Parallel-Axis Theorem:

* How do we "add" inertias if bolt two objects together?



What is J for the combined object.

- Look at total kinetic energy:

$$T = \frac{1}{2} m_1^N \underbrace{V_1^N V_1^N}_{\text{velocities of the CoM}} + \frac{1}{2} m_2^N V_2^N V_2^N + \frac{1}{2} \omega_1^B J_1^B \omega_1^B + \frac{1}{2} \omega_2^B J_2^B \omega_2^B$$

* Note we must align the body frames for both bodies

- If we connect the bodies, they must spin together

$$\Rightarrow \omega_1 = \omega_2 = \omega$$

- We'll look at pure rotation of the combined body

- By convention, $\mathcal{I}$ is defined at the CoM

$\Rightarrow$ Need to compute new CoM for combined body

- Write old CoM locations relative to new CoM as ${}^a\mathbf{r}_1$ and ${}^b\mathbf{r}_2$

$$\Rightarrow {}^a\mathbf{v}_1 = Q(\dot{\mathbf{r}}_1 + {}^a\boldsymbol{\omega} \times {}^a\mathbf{r}_1), \quad {}^b\mathbf{v}_2 = Q(\dot{\mathbf{r}}_2 + {}^b\boldsymbol{\omega} \times {}^b\mathbf{r}_2)$$

$$\Rightarrow T = \frac{1}{2} \boldsymbol{\omega}^T \mathcal{I}_1 \boldsymbol{\omega} + \frac{1}{2} \boldsymbol{\omega}^T \mathcal{I}_2 \boldsymbol{\omega}$$

$$+ \underbrace{\frac{1}{2} m_1 (\boldsymbol{\omega} \times {}^a\mathbf{r}_1)^T (\boldsymbol{\omega} \times {}^a\mathbf{r}_1)}_{\frac{1}{2} \boldsymbol{\omega}^T [m_1 S^2({}^a\mathbf{r}_1)] \boldsymbol{\omega}} + \underbrace{\frac{1}{2} m_2 (\boldsymbol{\omega} \times {}^b\mathbf{r}_2)^T (\boldsymbol{\omega} \times {}^b\mathbf{r}_2)}_{\frac{1}{2} \boldsymbol{\omega}^T [m_2 S^2({}^b\mathbf{r}_2)] \boldsymbol{\omega}}$$

- Combining:

$$T = \frac{1}{2} \boldsymbol{\omega}^T \underbrace{[\mathcal{I}_1 + \mathcal{I}_2 + m_1 S^2({}^a\mathbf{r}_1) + m_2 S^2({}^b\mathbf{r}_2)]}_{\mathcal{I}} \boldsymbol{\omega}$$

- Recipe:

- 1) Make sure body frames are aligned
- 2) Calculate new CoM
- 3) Add $\mathcal{I}$ matrices
- 4) Add point mass terms relative to new CoM

Momentum:

- For a particle, momentum $\vec{p} = m\vec{v}$

- Note that this is the gradient of kinetic energy!

$$T = \frac{1}{2} m \mathbf{v}^T \mathbf{v} \Rightarrow \frac{\partial T}{\partial \mathbf{v}} = m\mathbf{v} = \mathbf{p}$$

- The right way to think about this in general is:

$$p = \frac{\partial L}{\partial \dot{q}}$$

- For mechanical systems:

$$L = T - U = \frac{1}{2} \dot{q}^T M(q) \dot{q} - U(q)$$

$$\Rightarrow p = \frac{\partial L}{\partial \dot{q}} = \frac{\partial T}{\partial \dot{q}} = M(q) \dot{q}$$

* Conservation of momentum:

$$\underbrace{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right)}_{\dot{p}} - \frac{\partial L}{\partial q} = 0 \Rightarrow \dot{p} = \frac{\partial L}{\partial q}$$

$\Rightarrow$ if L doesn't depend on $q \Rightarrow$ corresponding $\dot{p} = 0$
 $\Rightarrow p$ is constant.

- There's a more general version of this idea (Noether's Theorem) that says there is a conserved quantity corresponding to every (continuous) symmetry of the system.

- For a rigid body, we have:

$$T = \frac{1}{2} m \tilde{v}^T \tilde{v} + \frac{1}{2} \omega^T \tilde{J} \omega$$

$$\Rightarrow \frac{\partial T}{\partial \tilde{v}} = \underbrace{m \tilde{v}}_{\text{linear momentum}}, \quad \frac{\partial T}{\partial \omega} = \underbrace{\tilde{J} \omega}_{\text{angular momentum}} = \tilde{h}$$

Euler's Equations:

- Translation and rotation decouple in the dynamics just like in kinetic energy (if we put origin of B at CoM)
- Analogy to Newton's 2nd Law:

$$\frac{d\tilde{\mathbf{p}}}{dt} = \tilde{\mathbf{F}}$$

↑ momentum
↑ force

$$\frac{d\tilde{\mathbf{h}}}{dt} = \tilde{\mathbf{\tau}}$$

↑ angular momentum
↑ torque

- We really want to write the angular part in the body frame so $\mathbf{J}$ is constant:

$${}^B\mathbf{h} = {}^B\mathbf{J} {}^B\boldsymbol{\omega} \Rightarrow {}^N\mathbf{h} = \mathbf{Q} {}^B\mathbf{J} {}^B\boldsymbol{\omega}$$

- Plug this back in:

$$\frac{d\tilde{\mathbf{h}}}{dt} = \frac{d}{dt}(\mathbf{Q} {}^B\mathbf{J} {}^B\boldsymbol{\omega}) = \underbrace{\dot{\mathbf{Q}}}_{\text{Q Speed}} \mathbf{J} {}^B\boldsymbol{\omega} + \mathbf{Q} \mathbf{J} \dot{{}^B\boldsymbol{\omega}} = \mathbf{Q}({}^B\boldsymbol{\omega} \times \mathbf{J} {}^B\boldsymbol{\omega} + \mathbf{J} \dot{{}^B\boldsymbol{\omega}}) = \tilde{\mathbf{\tau}}$$

- Multiply both sides by $\mathbf{Q}^T = {}^B\mathbf{Q}^N$

$$\Rightarrow \boxed{{}^B\mathbf{J} {}^B\dot{\boldsymbol{\omega}} + {}^B\boldsymbol{\omega} \times \mathbf{J} {}^B\boldsymbol{\omega} = {}^B\boldsymbol{\tau}} \leftarrow \text{"Euler's Equation"}$$

- The full (translation + rotation) dynamics of a rigid body are often called "Newton-Euler Dynamics":

$$\underbrace{\begin{bmatrix} m\mathbf{I} & \mathbf{0} \\ \mathbf{0} & {}^B\mathbf{J} \end{bmatrix}}_M \underbrace{\begin{bmatrix} {}^N\dot{\mathbf{v}} \\ {}^B\dot{\boldsymbol{\omega}} \end{bmatrix}}_{C(\mathbf{q}, \mathbf{v})} + \underbrace{\begin{bmatrix} \mathbf{0} \\ {}^B\boldsymbol{\omega} \times \mathbf{J} {}^B\boldsymbol{\omega} \end{bmatrix}}_{C(\mathbf{q}, \mathbf{v})} = \underbrace{\begin{bmatrix} {}^N\mathbf{F} \\ {}^B\boldsymbol{\tau} \end{bmatrix}}_{\text{Note Frames}}$$

- This is a standard model used in aircraft flight dynamics, spacecraft, underwater vehicles, legged robots, etc.

Free Rigid Body Motion:

- Look at just rotation with no external torque:

$$J\dot{\omega} + \omega \times J\omega = 0$$

- With no torque, momentum is conserved:

$${}^N\dot{h} = 0 \Rightarrow {}^N h = \text{const.} \Rightarrow \|{}^N h\| = \text{const.} \Rightarrow \|{}^B h\| = \text{const.}$$

$\Rightarrow {}^B h(t)$ lives on a sphere ("momentum sphere")

- We can re-write Euler's equation in terms of ${}^B h = J^B \omega$:

$${}^B\dot{h} + (J^{-1}h) \times h = 0$$

$$\Rightarrow {}^B\dot{h} = {}^B h \times J^{-1} {}^B h$$

- Energy is also conserved:

$$T = \frac{1}{2} \omega^T J \omega = \frac{1}{2} h^T J^{-1} h = \text{const.}$$

$\Rightarrow {}^B h(t)$ lives on an energy ellipsoid.

- * Combining these $\Rightarrow {}^B h(t)$ trajectories lie on intersections of momentum sphere and energy ellipsoid.

Equilibria:

$$\dot{h} = h \times J^{-1} h = 0 \Rightarrow \underbrace{h \parallel J^{-1} h}_{\text{"parallel"}} \Rightarrow J^{-1} h = \underbrace{\lambda h}_{\text{scalar}}$$

⇒ Equilibrium spins occur when $\mathbf{h}$ (or $\boldsymbol{\omega}$) is an eigenvector of $\mathbf{J}^{-1}$ (or $\mathbf{J}$)

- Define "Principle axes" as frame that diagonalizes $\mathbf{J}$:

$${}^P\mathbf{J} = \begin{bmatrix} J_{11} & 0 & 0 \\ 0 & J_{22} & 0 \\ 0 & 0 & J_{33} \end{bmatrix}, \quad {}^B\mathbf{J} = {}^B\mathbf{Q} {}^P\mathbf{J} {}^P\mathbf{Q}^B$$

eigenvectors of ${}^B\mathbf{J}$

- We'll assume $J_{11} < J_{22} < J_{33}$

minor intermediate major